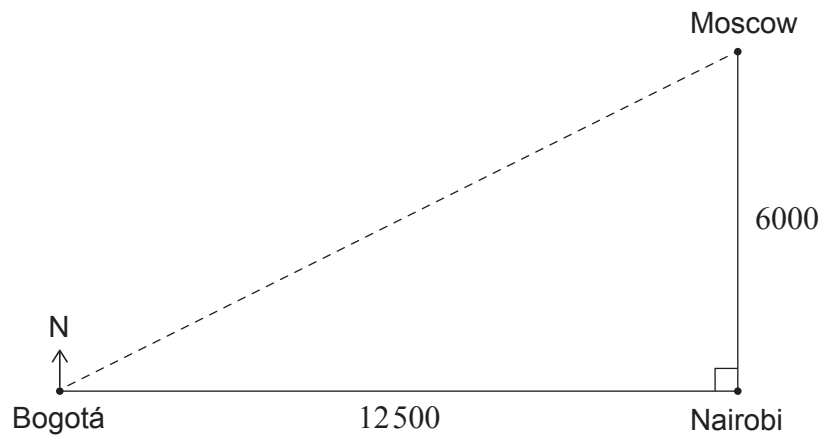


2. [Maximum mark: 27]

The following question compares the distance and direction between cities on a flat surface to the distance and direction between cities on a sphere.

Consider a model where the cities of Bogotá, Moscow, and Nairobi lie on a flat surface. In this model, Nairobi is 6000 km due south of Moscow and Bogotá is 12 500 km due west of Nairobi, as shown in the following diagram.

diagram not to scale



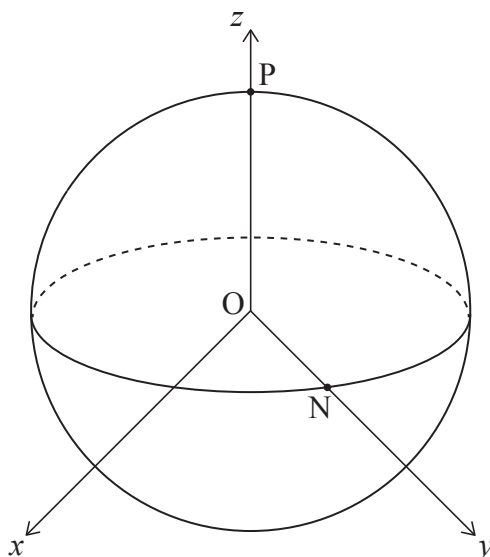
- (a) (i) Find the distance from Bogotá to Moscow. [2]
- (ii) Find the bearing of Moscow from Bogotá. Give your answer in degrees. [3]

(This question continues on the following page)

(Question 2 continued)

In reality, these three cities lie on the curved surface of the Earth which will change the distances and directions found in part (a).

Now consider a curved model using a coordinate system (x, y, z) with its origin, O , at the centre of the Earth. The units of this system are thousands of kilometres and the Earth is modelled as a sphere with radius 6000 km. The North Pole, P , lies on the z -axis, and Nairobi, N , is modelled as being on the equator and lying on the y -axis.



P has position vector $\vec{OP} = \mathbf{p} = \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix}$ and N has position vector $\vec{ON} = \mathbf{n} = \begin{pmatrix} 0 \\ 6 \\ 0 \end{pmatrix}$.

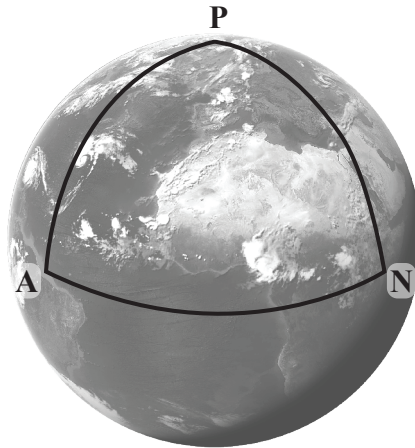
(b) (i) Use the scalar product to find the angle between $\mathbf{p}$ and $\mathbf{n}$. [2]

(ii) Show that the distance between P and N along the arc from P to N is 3000π km. [2]

(This question continues on the following page)

(Question 2 continued)

Point A , which is also on the equator, has position vector $\mathbf{a} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$ as shown in the following diagram.



P , N and A , and the arcs connecting them, form a spherical triangle.

The angle at vertex A is defined as the angle between the vectors $\mathbf{a} \times \mathbf{p}$ and $\mathbf{a} \times \mathbf{n}$.

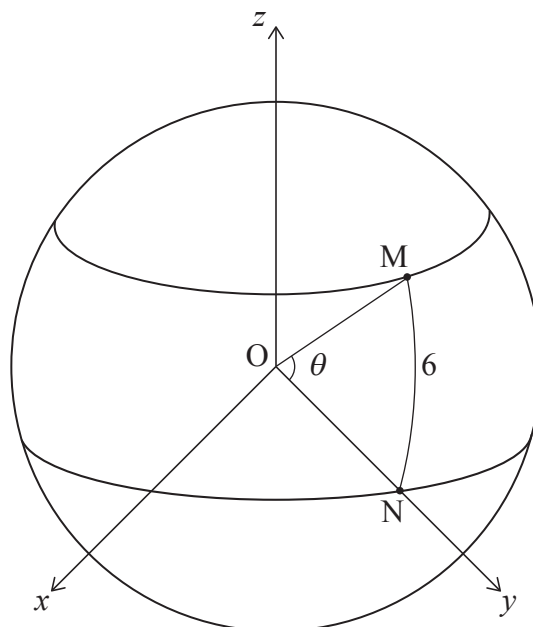
(c) (i) Find the vector $\mathbf{a} \times \mathbf{p}$. [2]

(ii) Show that the angle at vertex A in the spherical triangle is 90° . [3]

(This question continues on the following page)

(Question 2 continued)

Moscow, M, has position vector $\vec{OM} = \mathbf{m} = \begin{pmatrix} 0 \\ 6 \cos \theta \\ 6 \sin \theta \end{pmatrix}$, as shown in the following diagram.



The shortest distance between two points on the sphere lies along an arc of a circle on the sphere with centre O. In this model the shortest distance from Moscow to Nairobi is 6000 km.

- (d) Show that $\theta = 57.3^\circ$, correct to three significant figures. [2]

Bogotá, B, is west of Nairobi and has position vector $\vec{OB} = \mathbf{b} = \begin{pmatrix} 6 \sin 120^\circ \\ 6 \cos 120^\circ \\ 0 \end{pmatrix}$.

- (e) Find the shortest distance from Bogotá to Moscow on the sphere. [5]

The bearing from B to M is defined as the angle at vertex B in the spherical triangle

containing B, M and P. It is given that $\mathbf{b} \times \mathbf{p} = \begin{pmatrix} 36 \cos 120^\circ \\ -36 \sin 120^\circ \\ 0 \end{pmatrix}$.

- (f) Using the method from part (c), find the bearing from Bogotá to Moscow. [6]

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References:

- 2(c)** Google Maps/Google Earth. Data from Data SIO, NOAA, U.S. Navy, NGA, GEBCO Landsat / Copernicus, IBCAO, U.S. Geological Survey, PGC/NASA. Imagery from 14/12/2015. Image available at: https://earth.google.com/web/@12.01529518,-18.56070747,-158.39383184a,23597813.93249989d,30.00008083y,359.99981502h,0t,0r/data=CgRCaggBOgMKATBCAggASg0I_____ARAA.